

Sprint 3 Market Data

Delivered

Honest status: code, migrations, tests, and operational checks are delivered. Production pricing is inactive. Single-source sandbox pricing is not approved for real funds.

Goal and Connection

Sprint 1 created configuration-driven assets, fiat, networks, and markets. Sprint 2 added authentication, idempotency, audit, and durable work. Sprint 3 collects price facts and calculates safety-checked reference rates. Sprint 4 adds spreads, fees, limits, amounts, and expiry.

What Was Built

Part	Purpose
Domain	Exact direct and multi-leg math plus safety guards.
Adapters	Coinbase, manual, and deterministic providers behind interfaces.
PostgreSQL	Immutable evidence, routes, snapshots, and leased jobs.
Worker	Bounded claims, retries, and crash recovery.

End-to-End Example

These currencies are examples, not hardcoded scope.

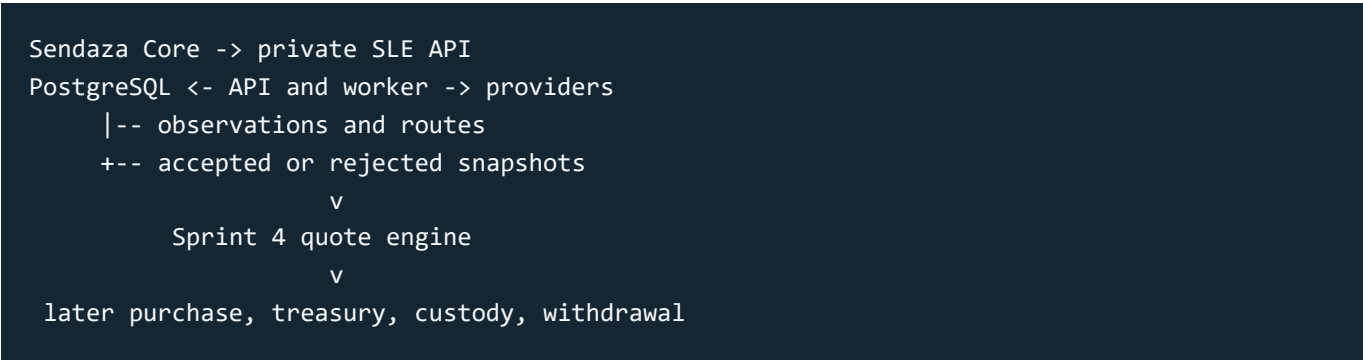
```
1 ETH = 2,500 USDT  provider fact
    | multiply
1 USDT = 1.00 USD   stablecoin guard
    | multiply
1 USD = 1,600 NGN   manual version
    v
1 ETH = 4,000,000.00 NGN snapshot
+ route version, times, and evidence links
    v
Sprint 4 adds spread, fee, limits, and expiry
```

The worker resolves server-controlled providers. Decimal strings are validated and stored. Ordered multiply or divide legs use `decimal.js`, normalize once with half-even rounding, run guards, and commit the snapshot plus evidence in one PostgreSQL transaction.

Safety Rules

- No floating point rate math.
- Missing, stale, deviating, sequence-gap, or depegged evidence rejects pricing.
- Accepted snapshots require one matching observation per leg.
- Evidence is immutable.
- Expired leases can be reclaimed after a worker crash.
- Provider timeouts never erase stored evidence.

Place in SLE



Sendaza owns customers, KYC, locks, ledger entries, and customer interfaces. Sprint 3 never changes customer balances, reserves inventory, or signs transactions.

Verification

Check	Result
Focused operations	5 suites, 36 tests passed
Full unit suite	31 suites, 158 tests passed
Railway <code>sle_test</code>	5 PostgreSQL tests passed
Static/build	TypeScript and ESLint passed; API and worker bundles compiled
Production	7 migrations; 0 enabled routes, jobs, or valid snapshots

The Nest optional checker reports `spawn EPERM` in the Windows sandbox. Bundles compile and standalone strict TypeScript passes.

Database Repair

Real PostgreSQL testing exposed a deferred snapshot-input guard defect. A shared-pool transaction and forward migration `20260901000300_snapshot_input_validation` added explicit expected, supplied, and matching input checks. It passed on `sle_test` before approved production deployment.

Limitations and Risks

- No production route or refresh job is active.
- Real funds require independent redundant sources.
- Coinbase reference data is not guaranteed liquidity or an executable price.
- Quotes, custody, treasury, network fees, withdrawals, finality, and reconciliation remain later work.
- The Railway project currently contains PostgreSQL; API and worker deployment is deferred.

Next Sprint

Sprint 4 builds immutable executable quotes from accepted snapshots. It adds spreads, fixed or percentage purchase fees, exact debit and asset amounts, limits, expiry, idempotency, and stable Sendaza-facing errors.

Glossary

Observation: one price fact. **Leg:** one route step. **Cross-rate:** a rate through intermediate pairs. **Snapshot:** a result linked to evidence. **Depeg:** movement outside a reference band. **Basis point:** 0.01 percent. **Lease:** temporary job ownership. **Idempotent:** repeatable without a second effect.